

MTH205 Analysis in one variable

Tutorial 8

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- (1) Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable with $f(a) = 0$. If there is $M > 0$ such that
- $$|f'(x)| \leq M|f(x)|$$
- for every $x \in [a, b]$ then show that $f \equiv 0$.
- (2) Let $g : [-1, 1] \rightarrow [-1, 1]$ be differentiable surjective map with $|g'(x)| \leq 1$ for $-1 \leq x \leq 1$. Then either $g(x) = x$ or $g(x) = -x$ for every x .
- (3) Give an example of a uniformly continuous function on $[0, 1]$ that is differentiable on $(0, 1)$ but whose derivative is not bounded on $(0, 1)$.
- (4) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.
- (5) If a rock is dropped from a height of 150 feet, its position t seconds after it is dropped until it hits the ground is given by the function $s(t) = -t^2 + 5t + 150$.
- (a) Determine how long it takes before the rock hits the ground.
- (b) Find the average velocity V_{avg} of the rock when it is released and when it hits the ground.
- (c) Find the time t when the instantaneous velocity of the rock is V_{avg} .